

I Year I Semester

PHYSICS FOR COMMUNICATION SYSTEMS**AI Integrated)**

Course Code	Year/Sem	L	T	Pr	P	Credits
26APH04T	I Year I Sem	3	0	0	0	3

Prerequisites: Engineering Mathematics, Basic Electronics, Electromagnetic Theory Fundamentals

Course Objectives

The objective of this course is to:

1. To provide knowledge on wave optics concepts such as interference, diffraction, and polarization with basic AI-assisted optical analysis applications.
2. To introduce lasers, fiber optics, and advanced sensing technologies with simple AI-assisted communication and signal processing techniques.
3. To equip students the fundamentals of quantum mechanics and quantum computing with basic AI-assisted secure communication concepts.
4. To familiarize students with electrical properties of metals and semiconductors using AI-assisted material and conductivity analysis methods.
5. To impart knowledge on electromagnetic field theory and its applications with basic AI-assisted antenna and RFID system concepts.

Course Content**UNIT I: Wave Optics and Quantum Information Processing****9 hrs****Core Content:**

Interference: Principle of superposition and coherence, Interference, Types of interference, Interference in thin films (reflection geometry) and Newton's rings. Applications: Determination of wavelength and refractive index.

Diffraction: Introduction to diffraction, Fresnel and Fraunhofer diffraction, Fraunhofer diffraction due to single, double and N-slits (Diffraction Grating), Dispersive power and Resolving power of grating. Applications: CD/DVD data storage, Barcode scanners.

Polarization: Types of polarization, Nicol's prism, Half-wave and Quarter-wave plates.

AI Integration:

AI-assisted polarization management for quantum key distribution. AI-integrated holographic interferometry for surface flatness analysis.

Free/Open-Source AI Tools: Python (NumPy, SciPy, Matplotlib and scikit-learn).

UNIT II: Quantum Photonics and Advanced Sensing**9 hrs****Core Content:**

Lasers: Interaction of radiation with matter, Requisites and conditions for laser action, He-Ne laser, Nd-YAG laser, LiDAR basics. Laser-based data communication, Free-space optical communication, Optical data storage, Photonic computing basics, Applications of lasers.

Fiber Optics: Propagation mechanism, Numerical aperture, Acceptance angle, Types of optical fibres, Classifications and Modes of optical fibres, Attenuation and Losses in optical fibers, V-Number - Distributed Fiber Optic Sensing (DFOS) with AI-Powered Signal Analysis.

AI Integration:

AI-powered signal analysis for distributed fiber optic sensing. Python simulation of laser propagation and fiber optic signal attenuation.

Free/Open-Source AI Tools: Google Colab, Python, NumPy, SciPy and Scikit-learn/ TensorFlow.

UNIT III: Introduction to Quantum Mechanics & Quantum Computing**10 hrs****Core Content:**

Introduction to quantum mechanics, de-Broglie concept (matter waves), Heisenberg's Uncertainty Principle, Wave function, Postulates of Quantum Mechanics, Schrödinger's time-independent wave equation, Particle in a one-dimensional potential well, Quantum tunnelling (basic idea).

Introduction to Quantum Computing: Classical bits vs Qubits. Concept of superposition and Entanglement, Basic Quantum Operations: Hadamard gate (idea of superposition), Pauli-X gate. Basic idea of quantum measurement. Applications (qualitative only): AI-Assisted Tunnel Diodes and cold emission of electrons, Basic idea of quantum cryptography and secure communication.

AI Integration: AI-Assisted Analysis of Tunnel Diode Characteristics using Quantum Tunnelling Concepts, AI-Assisted Quantum Key Distribution for Secure Communication

Free/Open-Source AI Tools: Google Colab, Python and Scikit-learn.

UNIT IV: Electrical Properties of Metals and Semiconductors**8 hrs****Core Content:**

Classical free electron theory – merits and demerits; Quantum free electron theory – equation for electrical conductivity, Fermi-Dirac statistics, Fermi level and variation with temperature, Classification of solids – metals, insulators and semiconductors.

Semiconductors: Introduction, Fermi energy and Fermi level in intrinsic and extrinsic semiconductors, Expression for concentration of electrons in conduction band and holes in valence band, Electrical conductivity, Expressions for drift and diffusion currents, Hall effect, Expression for Hall coefficient and its applications

AI Integration: AI-Assisted Prediction of Electrical Conductivity in Metals and Semiconductors, AI-Assisted Carrier Concentration and Conductivity Analysis in Semiconductors.

Free/Open-Source AI Tools: Google Colab, Python and Scikit-learn.

UNIT V: Introduction to EM Theory**9 hrs****Core Content:**

Introduction, wave equation in differential form in free space, Vector analysis, Divergence and curl of electric field and magnetic field (static), Gauss' divergence theorem and Stokes' theorem. Derivations of the four Maxwell equations and their significance. Maxwell's equations in vacuum, Energy transport and Poynting Vector. Applications: Radio Frequency Identification (RFIDs) and Antennas.

AI Integration: AI-Assisted RFID-Based Smart Tracking and Identification Systems, AI-Assisted Fault Detection in Antenna Systems,

Free/Open-Source AI Tools: Google Colab, Python and Scikit-learn.

Course Outcomes (COs)

After completion of the course, student will be able to

CO	Course Outcomes	Bloom's Level
CO1	Apply the principles of interference, diffraction, polarization, along with basic AI-assisted optical analysis in engineering applications.	L4 (Analyze)
CO2	Analyze the working of lasers, fiber optics, and advanced sensing systems along with basic AI-assisted optical communication and signal processing techniques..	L4 (Analyze) L5 (Evaluate)
CO3	Apply the concepts of quantum mechanics and quantum computing along with basic AI-assisted secure communication applications.	L3 (Apply) L4 (Analyze)
CO4	Analyze the electrical properties of metals and semiconductors, along with basic AI-assisted material characterization and conductivity analysis methods	L4 (Analyze) L5 (Evaluate)
CO5	Apply Maxwell's equations and electromagnetic field concepts along with AI-assisted RFID and antenna-based systems.	L3 (Apply) L4 (Analyze)

CO – PO Articulation Matrix

Scale: 3 – Strong, 2 – Moderate, 1 – Slight, – No Correlation

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	2	2	–	–	–	–	–	–
CO2	3	2	2	2	3	–	–	–	–	–	–
CO3	3	3	2	2	2	–	–	–	–	–	–
CO4	3	3	2	2	2	–	–	–	–	–	–
CO5	3	3	2	2	3	–	–	–	–	–	–

Textbooks

1. Avadhanulu, Kshirsagar & Arun Murthy, A Textbook of Engineering Physics, S. Chand, 2019.
2. D. K. Bhattacharya & Poonam Tandon, Engineering Physics, Oxford Press, 2015.

Reference Books

3. B. K. Pandey & S. Chaturvedi, Engineering Physics, Cengage Learning, 2021.
4. Shatendra Sharma & Jyotsna Sharma, Engineering Physics, Pearson, 2018.
5. D. J. Griffiths, Introduction to Quantum Mechanics, Cambridge Press.
6. Saleh & Teich, Fundamentals of Photonics, Wiley.
7. Streetman & Banerjee, Solid State Electronic Devices, Pearson.
8. C. Kittel, Introduction to Solid State Physics, Wiley.

Web Resources & NPTEL Links

- NPTEL – nptel.ac.in
- MIT OpenCourseWare – ocw.mit.edu
- HyperPhysics – hyperphysics.phy-astr.gsu.edu
- Feynman Lectures – feynmanlectures.caltech.edu
- PhET Simulations – phet.colorado.edu

Recommended Free & Open-Source AI/Computational Tools

- Python (NumPy, SciPy, OpenCV, Matplotlib) – FREE on Google Colab for optics and EM simulations
- IBM Qiskit (free, open-source) – For quantum computing and quantum cryptography simulations
- scikit-learn (free, open-source) – For semiconductor data analysis and signal classification
- OpenEMS (free, open-source) – For electromagnetic field simulation and antenna design
- PhET Interactive Simulations (free) – For wave optics, semiconductors and quantum phenomena

I Year – I Semester

Course Code	PHYSICS FOR COMMUNICATION SYSTEMS LAB	L	T	Pr	P	Credits
26APH04P		0	0	0	3	1.5

Course Objectives

1. To determine optical properties—including dispersive power, thin-film thickness, radius of curvature, and spectral wavelengths—using prism spectroscopy, interference patterns, and diffraction gratings.
2. To characterize laser light transmission and optical waveguide parameters by measuring laser wavelengths through diffraction and calculating the numerical aperture of optical fibers.
3. To investigate atomic physics principles by estimating Planck's constant via the photoelectric effect and calculating the specific charge (e/m) of an electron using J.J. Thomson's method.
4. To analyze the electronic and dielectric properties of materials—including ferroelectric Curie temperature, dielectric constant, semiconductor band gap, resistivity, and Hall coefficient.
5. To evaluate magnetic and electromagnetic phenomena by analyzing B–H hysteresis curves, verifying the magnetic inverse square law with a Hall probe, and measuring coil self-inductance using LCR and resonance techniques.

List of Experiments (Any ten experiments need to be performed)

Part A – Regular Experiments

1. Determination of the dispersive power of the prism
2. Determination of the thickness of a thin object by the wedge method
3. Determination of the radius of curvature of the lens by Newton's rings
4. Determination of wavelengths of various colours of mercury spectrum using diffraction grating in the normal incidence method
5. Determination of wavelength of a LASER using diffraction grating
6. Determination of Numerical aperture of an optical fiber.
7. Estimation of Planck's constant using the photoelectric effect
8. Determination of e/m of an electron by J.J. Thomson method.
9. Determination of the Curie temperature and dielectric constant of a ferroelectric material
10. Study the variation of B versus H by magnetizing the magnetic material (B-H curve)
11. Determination of the resistivity of a semiconductor by the four-probe method.
12. Determination of the energy gap of a semiconductor using p-n junction diode
13. Determination of Hall voltage and Hall coefficient of a given semiconductor using Hall effect.
14. Verification of the inverse square law for a magnetic field using a Hall probe
15. Measurement of self-inductance of a coil using LCR meter and the resonance method

Part B – AI Integration Lab Exercises

Any **three** of the following AI exercises must be completed using Python / Google Colab or appropriate AI/LLM tools.

1. Optics Data Automation (Python / Matplotlib): input raw angle-of-deviation data from the prism experiment; fit a polynomial to the deviation vs. angle curve; extract the angle of minimum deviation and compute dispersive power; overlay the fitted curve on experimental data points.
2. Fringe Analysis using OpenCV (Python / Google Colab): import Newton's ring or wedge-fringe images; apply Gaussian blur and Canny edge detection to locate fringe centres; compute fringe diameters automatically; calculate radius of curvature or thickness and compare with manual measurements.
3. Planck's Constant Estimation (Python / NumPy): input stopping potential vs. frequency data from the photoelectric experiment; apply linear regression (V_s vs. ν); extract slope to compute h and y-intercept to find work function; plot the regression line and quantify percentage error against the accepted value 6.626×10^{-34} J·s.

AI Integration:

AI integration adds three exercises after the regular experiments: (i) Python automation of prism deviation curve fitting and dispersive power extraction using Matplotlib/NumPy; (ii) OpenCV-based fringe image analysis for Newton's rings and wedge method — automating diameter measurement and radius of curvature computation; (iii) linear regression on photoelectric effect data to extract Planck's constant and work function — reinforcing the connection between the physical experiment and quantum theory.

Free / Open-Source AI Tools:

Python (NumPy, SciPy, OpenCV, Matplotlib) – Google Colab (free); scikit-learn (free) – semiconductor data analysis; PhET Interactive Simulations (free) – wave optics and quantum phenomena; VLabs IIT (free) – online physics experiments.

Course Outcomes (COs)

After successful completion of this lab course, the student will be able to:

CO	Course Outcomes	Bloom's Taxonomy Level
CO1	Compute optical constants, surface curvatures, film thicknesses, and light wavelengths using interference, diffraction, and prism spectroscopy techniques	L3 (Apply)
CO2	Analyze light propagation dynamics to determine laser wavelengths and evaluate light acceptance angles/numerical aperture in optical fibers	L3–L4 (Apply, Analyse)
CO3	Analyze fundamental physical constants (h and e/m) to validate classical and quantum electron/photon interaction models	L4 (Analyse)
CO4	Evaluate the electronic, dielectric, and transport properties of semiconductor and ferroelectric materials across varying temperature and magnetic conditions	L4–L5 (Analyse, Evaluate)
CO5	Evaluate magnetic field profiles, material magnetization hysteresis (B–H), and inductive circuit resonance parameters against theoretical electromagnetic laws	L5–L6 (Evaluate, Create)

CO – PO Articulation Matrix (NBA)

Scale: 3 – Strong | 2 – Moderate | 1 – Slight | – No Correlation

CO	Bloom's Lvl	PO1 Eng. Know.	PO2 Prob. Analysis	PO3 Design/ Dev.	PO4 Invest- igation	PO5 Modern Tools	PO6 Eng. & Society	PO7 Env. & Sust.	PO8 Ethics	PO9 Team Work	PO10 Comm un.	PO11 Prj Mgmt
CO1	L3	3	2	2	3	2	–	–	–	–	–	–
CO2	L3–L4	3	3	3	3	3	–	–	–	–	–	–
CO3	L4	3	3	2	3	2	–	–	–	–	–	–
CO4	L4–L5	3	3	3	3	3	–	–	–	–	–	–
CO5	L5–L6	2	3	3	3	3	–	–	–	–	–	–

Recommended Free & Open-Source AI / Computational Tools

Python (NumPy, SciPy, OpenCV, Matplotlib) on Google Colab (free) – optics automation, fringe analysis, photoelectric regression, EM simulations

scikit-learn (free, open-source) – semiconductor data analysis and signal classification

OpenEMS (free, open-source) – electromagnetic field simulation and antenna design

PhET Interactive Simulations (free) – wave optics, semiconductors, and quantum phenomena: phet.colorado.edu

Virtual Labs – IIT (free): https://iitb.vlabs.co.in/discipline.html?discipline=Physical_Sciences – online simulation of optics, semiconductor and EM experiments

HyperPhysics (free) – concept reference for optics, quantum and EM theory: hyperphysics.phy-astr.gsu.edu

LibreOffice Calc (free, open-source) – standardised data recording templates for all experiments

Note: Lab record to be submitted for every experiment at the next session. AI exercises to be submitted as Google Colab .ipynb notebooks.

Reference Books

1. M. N. Avadhanulu & P. G. Kshirsagar, *A Textbook of Engineering Physics*, S. Chand & Company, 2019.
2. H. K. Malik & A. K. Singh, *Engineering Physics*, Tata McGraw-Hill, 2010.
3. R. K. Gaur & S. L. Gupta, *Engineering Physics*, Dhanpat Rai Publications, 2012, 8th Edition.
4. B. K. Pandey & S. Chaturvedi, *Engineering Physics*, Cengage Learning, 2012.